Multiplayer Game Platform — High-Level Architecture Overview

Developed by Bogo Solutions

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1. System Purpose

The **Multiplayer Game Platform** is a turn-based online multiplayer game platform developed by Bogo Solutions. It provides shared infrastructure—authentication, matchmaking, game session management, turn handling, player profiles, leaderboards, and notifications—so that any conforming game (Tic-Tac-Toe, Connect Four, etc.) can be hosted and played without rebuilding platform-level concerns.

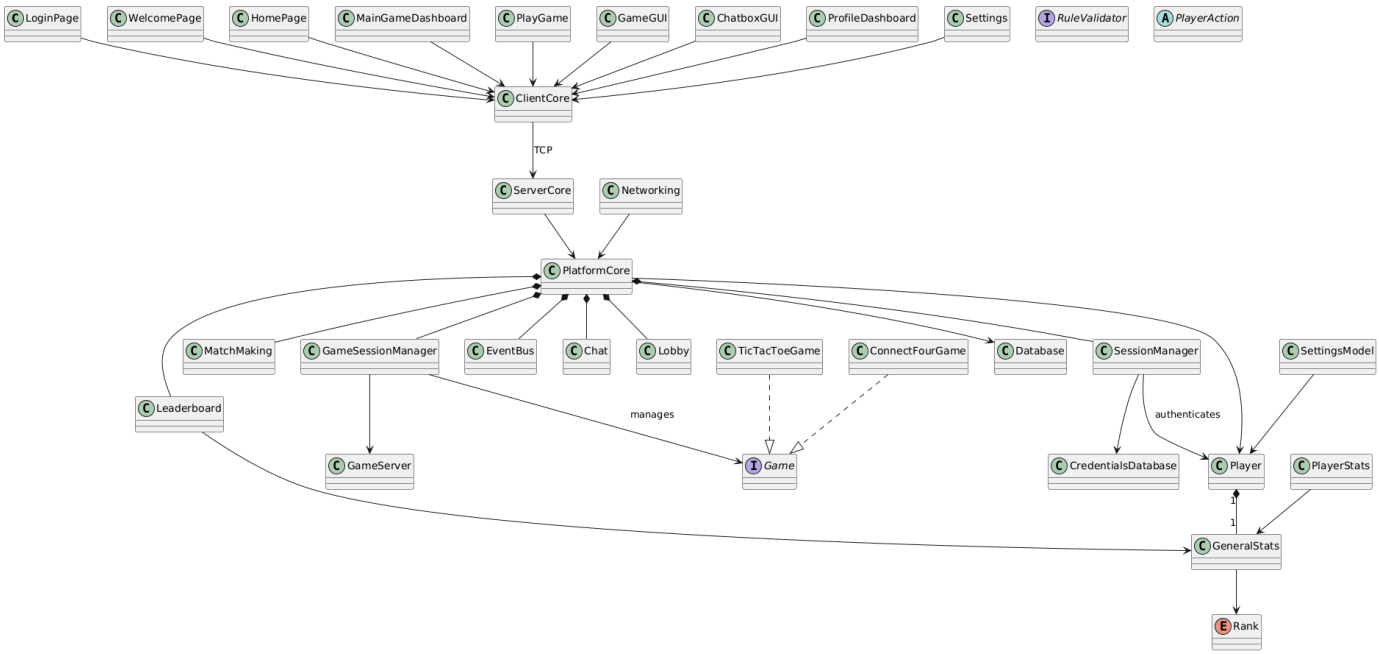
> **Platform-first principle:** The platform is the product. Sample games exist only to demonstrate and validate

> platform functionality. Every architectural decision prioritises shared, reusable services over game-specific logic.

2. Architectural Principles

Principle	Description
Platform owns notifications	`GameSessionManager` emits `GameStateUpdate` events; `PlatformCore` decides who to notify and how, regardless of where the player currently is (lobby, dashboard, or in-game).
GameEngine returns state only	`Game` implementations return state (or deltas). They do NOT notify the lobby, dashboard, or UI. Core services consume returned state and emit `GameStateUpdate`.
UI calls PlatformCore APIs	The UI layer calls `PlatformCore` APIs (`submitMove`, `joinQueue`, `joinLobby`). It never calls `GameEngine` internals.
Stubs for external systems	Database, networking/server, and cloud hosting are represented by stub classes. Architecture is designed so a real implementation can be swapped in at the stub boundary without structural changes.
Event-driven decoupling	`GameStateUpdate` is the contract between game logic and everything else. No direct calls from game layer to UI.
Privacy by design	`Lobby.players` is encapsulated; external code cannot enumerate who is in a lobby.
Single responsibility	Each service owns one lane: matchmaking, sessions, lobby, leaderboard, or chat.

3. Top-Level Component Map



4. Layer Breakdown

4.1 UI & Navigation Layer

The UI layer is the player-facing surface. It is passive: it **subscribes** to `PlatformCore` notifications and renders state; it never drives game logic.

Class	Responsibility
`WelcomePage`	Entry point; displays welcome message
`AuthenticationScreen`	Displays login and registration screens; calls `LoginPage.login()` / `register()`
`LoginPage`	Delegates login/register to `PlatformCore`; returns `HomePage` or error `State`
`HomePage`	Player home; entry to profile, settings, and game selection
`MainGameDashboard`	Central hub; `displayGames`, `chooseGame`, `viewProfile`, `showLeaderboard`; joins queue; receives state updates and turn alerts from `PlatformCore`
`ProfileDashboard`	Displays profile view and per-game stats
`Settings`	Exposes account management (`changeUsername`, `changeEmail`, `changePassword`, `logout`)
`PlayGame`	Manages active game view; calls `PlatformCore.submitMove()`; receives board updates
`GameGUI`	Renders the game board (`setupUIWindow`, `renderBoard`, `updateUI`)

Class	Responsibility
`ChatboxGUI`	In-game chat display
`State` (enum)	Login/register error states: `UsernameTaken`, `EmailFormatWrong`, `VerificationCodeWrong`

Navigation flow: `WelcomePage → AuthenticationScreen → LoginPage → HomePage → MainGameDashboard ↔ PlayGame`

4.2 Platform Core & Services

This is the central nervous system of the platform. All game events and player notifications are routed through here. The architecture is split across a **client-side** and a **server-side** separated by a TCP network stub. The `ClientCore` handles all local requests and client-side communication; `ServerCore` is the main functional component on the server side and routes all traffic between services.

> **Class diagrams:** See §10.1.1 (rough design), §10.1.2 (simplified, for other sub-teams), and §10.1.3 (detailed).
> Images: `02_01_Platform_Core_and_Services.png`, `2_01_Platform_Core_Simplified.png`, `2_01PlatformCore_drawio_2_.png`

Client-Side Components

`ClientUI` ***(interface)***

The full UI & navigation layer (owned by the Client/UI sub-team). All user interaction flows through this interface.

- `send(request)` — sends a user action to `ClientCore`
- `receive(update)` — receives updates from `ClientCore` and renders them

`clientGame` ***(interface)***

Client-side instance of the game validation logic (owned by the Rules & Validation sub-team). `ClientCore` uses this to do a lightweight local check before sending a move to the server, reducing unnecessary network round trips.

- `validate(move): Bool` — returns whether a move is locally valid

`ClientCore` ***(service)***

Handles all local requests from the user, communicates with `ServerCore` for remote requests, stores local information for the UI, and checks moves are valid before sending to the server.

- `send(message): String` — serialises and sends a request over `NETWORK`
- `receive(String): message` — receives and deserialises server responses
- `check(move)` — calls `clientGame.validate(move)` before forwarding to server

> **Note:** The exact distribution of functions between the main program thread and the listener thread will be determined when coding starts. This split is not final.

`ClientReceiver` ***(service)***

Listens for incoming messages from the server, determines the type of update, and acts on it.

- `receiveMessage(message): getUpdate`
- `getUpdate(message): update`
- `sendUpdate(update): void`

Network Layer

`NETWORK` ***(stub)***

Messages are sent and received over the network by TCP. For communicating between threads and different programs, using actual networking (same IP, different ports) is the chosen solution. The `NETWORK` stub is the boundary point between client and server.

> Dotted lines from `Chat`, `GameLobby`, and `Game` back to the client represent events sent via `ServerCore` to the user; these are separated out in the diagram for clarity.

Server-Side Components

`ServerCore` ***(service)***

The main functional component of Platform Core. All communication between services is handled here. `ServerCore` receives messages from any client, routes user actions to the correct service, hosts the 'main' lobby for global chat, and handles new connections.

- `send(ID, message): String` — sends a message to a specific client
- `receive(ID, String): ID, message` — receives and routes an incoming client message
- `routeContent(content): void` — dispatches content to the appropriate service

`Database` ***(stub)***

Stub for all persistence. Returns login boolean, handles scoreboard updating, and provides the interface that a real database implementation will conform to.

- `storeData(): void`
- `retrieveData(): void`

`GameLobby` ***(service)***

Provides a separate chat and UI context from the main lobby. Displays game info and handles the entry point to matchmaking.

- `receiveContent(content): void`
- `requestJoin()` — triggers the matchmaking flow

`matchmaking` ***(service)***

Puts users in the queue. When a match is found, routes the join request via `GameLobby`; if accepted, creates a new `Game` instance and moves both players into it.

`Chat` ***(service)***

Sends chat messages to all users on its list. Maintains `List<int> users` (user IDs) and `List<int>` user IP addresses.

- `receiveContent(content)` → `sendMessage(content, users)` — broadcasts content to the channel
- `sendMessage(content, users)` — delivers the chat message to everyone on the list

`Game` ***(service)***

Sends and receives turn updates, checks the validity of moves, sends the updated game state to both players, provides a local chat instance, and allows users to return to the lobby or quit.

`GameEngine` ***(interface)***

The rules and validation contract. `Game` delegates all move legality and state-transition logic to this interface.

- `validateMove(move): Boolean`
- `updateGame(move)` — returns the new game state and next turn value

`GameSessionManager` ***(service)***

Owns the move pipeline. Calls `Game` interface methods; emits `GameStateUpdate` on move accepted/rejected, turn changes, and game end.

Method	Behaviour
`startSession(roomId, gameType)`	Creates a game session; returns `sessionId`
`submitMove(sessionId, playerId, move)`	Calls `Game.validateAction` → `handleAction` → `isGameOver`; publishes `GameStateUpdate`
`endSession(sessionId)`	Tears down the session

`EventBus` ***(service)***

Simple pub/sub bus. `GameSessionManager` publishes `GameStateUpdate` events here; `PlatformCore` subscribes.

`GameStateUpdate` ***(event)***

The canonical message passed between game layer and platform layer.

Field	Type	Description
`sessionId`	String	Identifies the game session
`recipients`	List<String>	Player IDs who should receive this update
`status`	String	`YOUR_TURN` , `WAITING` , `ENDED`
`activePlayerId`	String	Which player acts next
`payload`	Object	Board snapshot or delta

`GameStatus` ***(enum)***

`LOBBY → MATCHMAKING → STARTING → IN_PROGRESS → GAME_OVER`

`MatchMaking` ***(service)***

Handles ranked queue management by `gameType` with skill-based player matching.

- `addPlayerToQueue(playerId, gameType)` — adds player to ranked queue
- `removePlayer(playerId)` — dequeues player
- `matchPlayers(gameType): List<String>` — returns matched player IDs; triggers `PlatformCore` when match is found

`Leaderboard` ***(service)***

- `logGameResult(sessionId)` — records result after game ends
- `updatePlayer(playerId)` — recalculates rank/MMR
- `getTopPlayersByRank(gameType): List<String>` — returns ranked player list

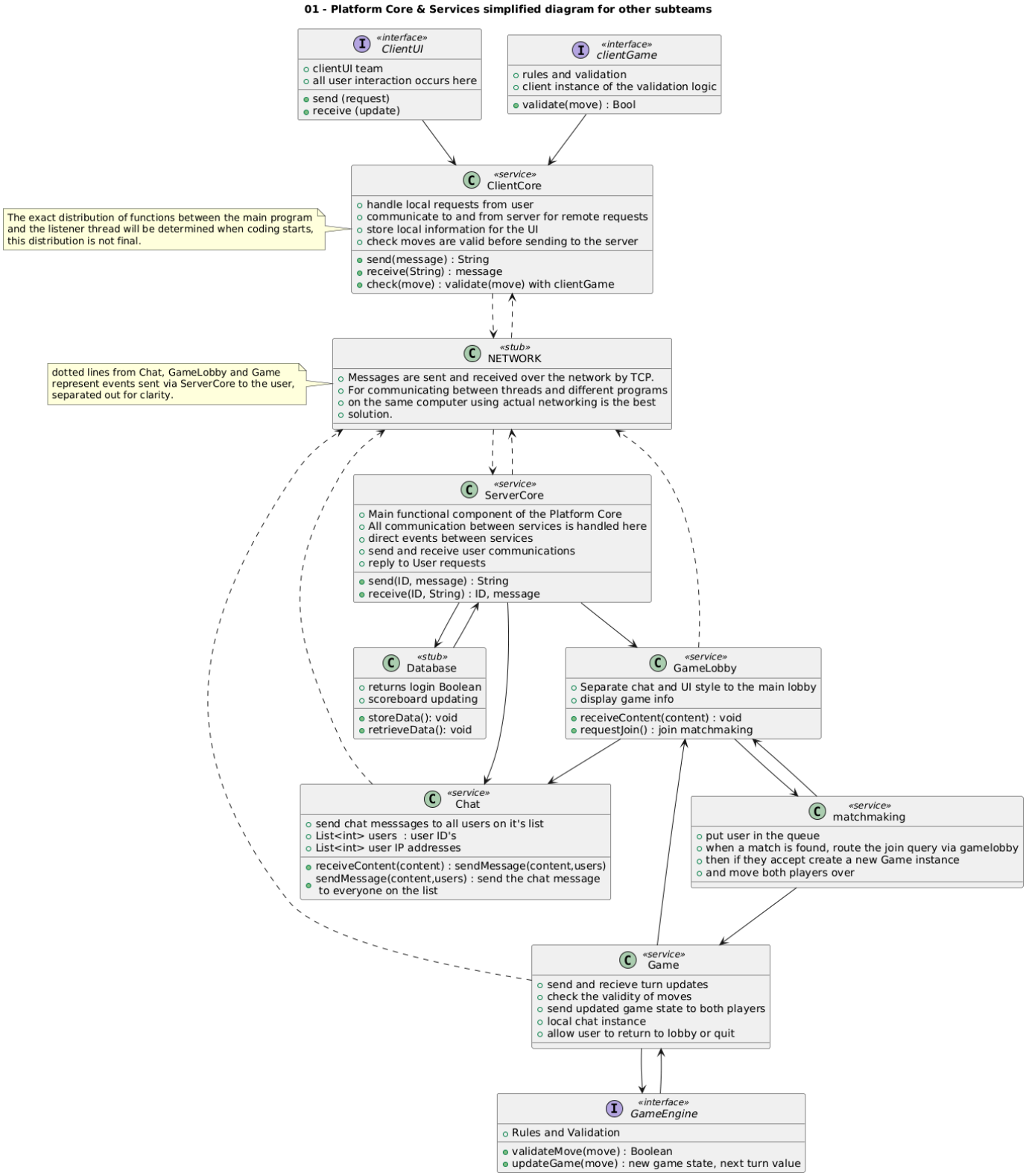
Backed by `Database` (stub).

`SessionManager`

Manages authentication sessions: creates session tokens on login, invalidates them on logout, stores session data locally.

Platform Core — Class Diagrams

Simplified Design (for other sub-teams)



4.3 Game Engine Layer

The game layer defines a shared abstraction that allows multiple games to run on the same platform infrastructure while sharing networking, session management, and validation logic. The platform interacts only with these abstractions and does not depend on any game-specific logic.

`Game` ***(interface)***

Represents a single running match of any game type.

Responsibilities:

- Initialize and start a game session
- Accept player actions
- Validate actions before applying them
- Maintain and expose game state
- Send just the changes (updates) instead of whole game state to the client
- Determine when the game has ended

Method	Description
`startGame()`	Initializes and starts the game session
`handleAction(PlayerAction action)`	Accepts and processes a player action
`validateAction(PlayerAction action)`	Validates an action before it is applied
`getGameState()`	Returns the current full game state
`getGameUpdate()`	Returns only the changes since last update
`isGameOver()`	Returns whether the game has ended

`PlayerAction` ***(abstraction)***

All player inputs are represented as a generic action object interpreted by the active game. The platform does not interpret the contents of actions; it only routes them to the correct game instance.

Field	Description
Player ID	Identifies which player submitted the action
Action type	e.g., `MOVE` , `HIT` , `STAND`
Action payload	Game-specific data; format differs between games

Each game provides its own concrete implementation of `PlayerAction` .

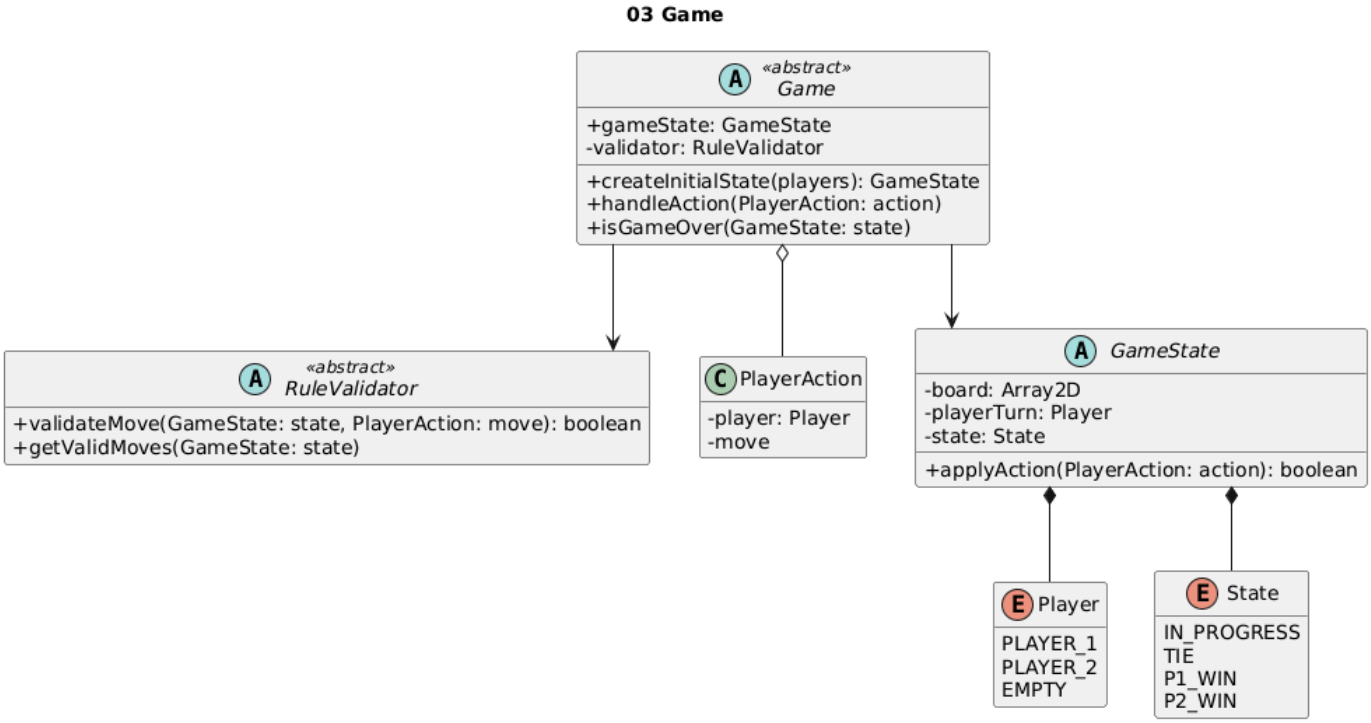
`RuleValidator` ***(interface)***

Validation logic is separated from game flow logic. Each game provides its own implementation.

Responsibilities:

- Determine whether a proposed action is legal
- Enforce turn order and game-specific constraints
- Prevent invalid state transitions
- Upon validation failure, return a reason why it failed rather than a mere boolean
- Provide a list of valid moves to the client

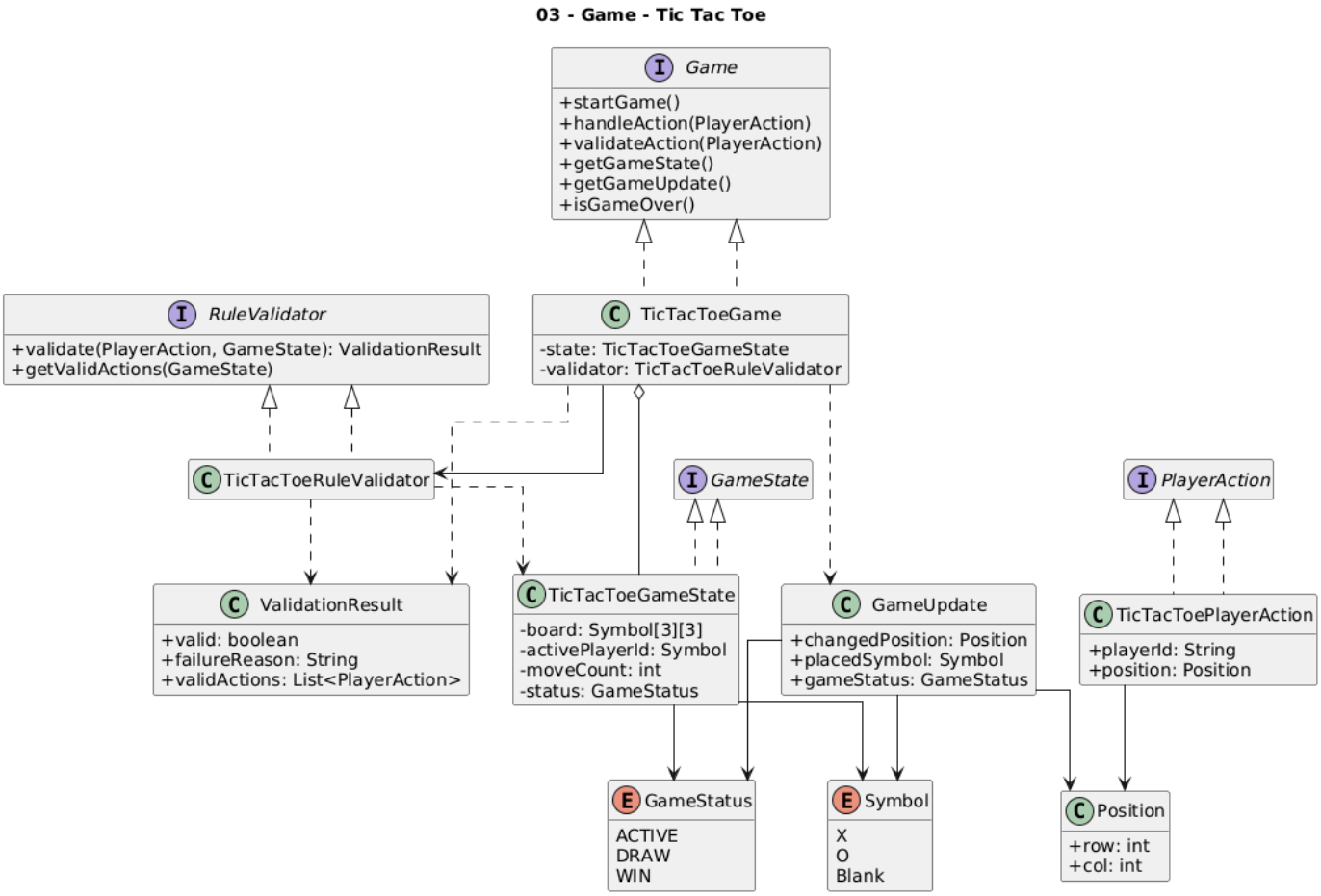
Method	Description
`validate(PlayerAction action, GameState state)`	Returns validation result with failure reason if invalid
`getValidActions()`	Returns the list of currently legal actions



Tic-Tac-Toe Implementation

Components:

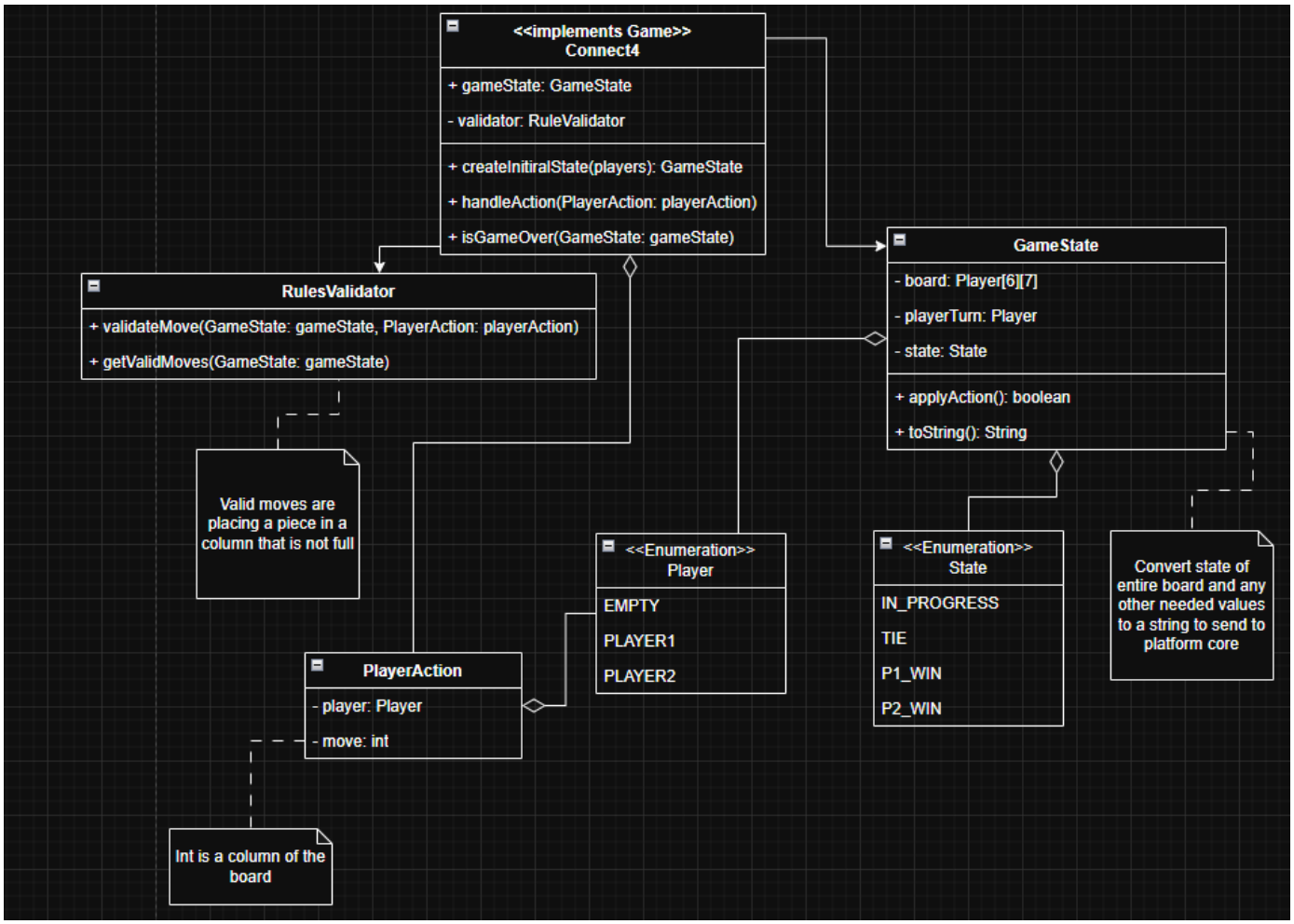
- `TicTacToeGame` (implements `Game`)
- `TicTacToePlayerAction` (concrete `PlayerAction`)
- `TicTacToeRuleValidator` (implements `RuleValidator`)
- `TicTacToeGameState` (board positions, current player)



Connect Four Implementation

Components:

- `ConnectFourGame` (implements `Game`)
- `ConnectFourPlayerAction` (concrete `PlayerAction`)
- `ConnectFourRuleValidator` (implements `RuleValidator`)
- `ConnectFourGameState` (board positions, current player)



Interaction Flow (Game Layer)

1. Client sends an action to the server
2. Platform routes the action to the active `Game` instance
3. `Game` requests validation from its `RuleValidator`
4. If valid, the game state is updated
5. Updated state is returned to the platform for broadcasting

> **Rule:** Game implementations never call UI or platform notification methods. They only return state objects to `GameSessionManager`.

4.4 Player, Profiles, Stats & Persistence

`Player`

Central domain object representing an authenticated user.

Member	Description
`playerId: String`	Unique identifier

Member	Description
`username: String`	Display name
`email: String`	Contact / login credential
`password: String`	Stored as bcrypt hash
`ranking: Rank`	Current overall rank
`getStats(gameType): GeneralStats`	Per-game statistics
`login()` / `logout()`	Session hooks

`GeneralStats`

Per-player, per-game statistics: `gamesPlayed`, `wins`, `losses`, `ties`, `mmr`, `rank`. Exposes `win()`, `lose()`, `tie()`, `updateMMR(win)`, `updateRank()`.

`Rank` ***(enum)***

`BRONZE → SILVER → GOLD → PLATINUM → DIAMOND → MASTER → GRANDMASTER`

`PlayerStats`

Convenience wrapper providing game-specific stat getters (`getWinsForConnect4()`, `getRankForTicTacToe()`, etc.) by delegating to `GeneralStats`.

`Settings`

Account management operations (change username/email/password, delete account, logout). References both `Player` and `CredentialsDatabase`.

`CredentialsDatabase` ***(stub)***

In-memory `HashMap<String, Player>` store. Provides:

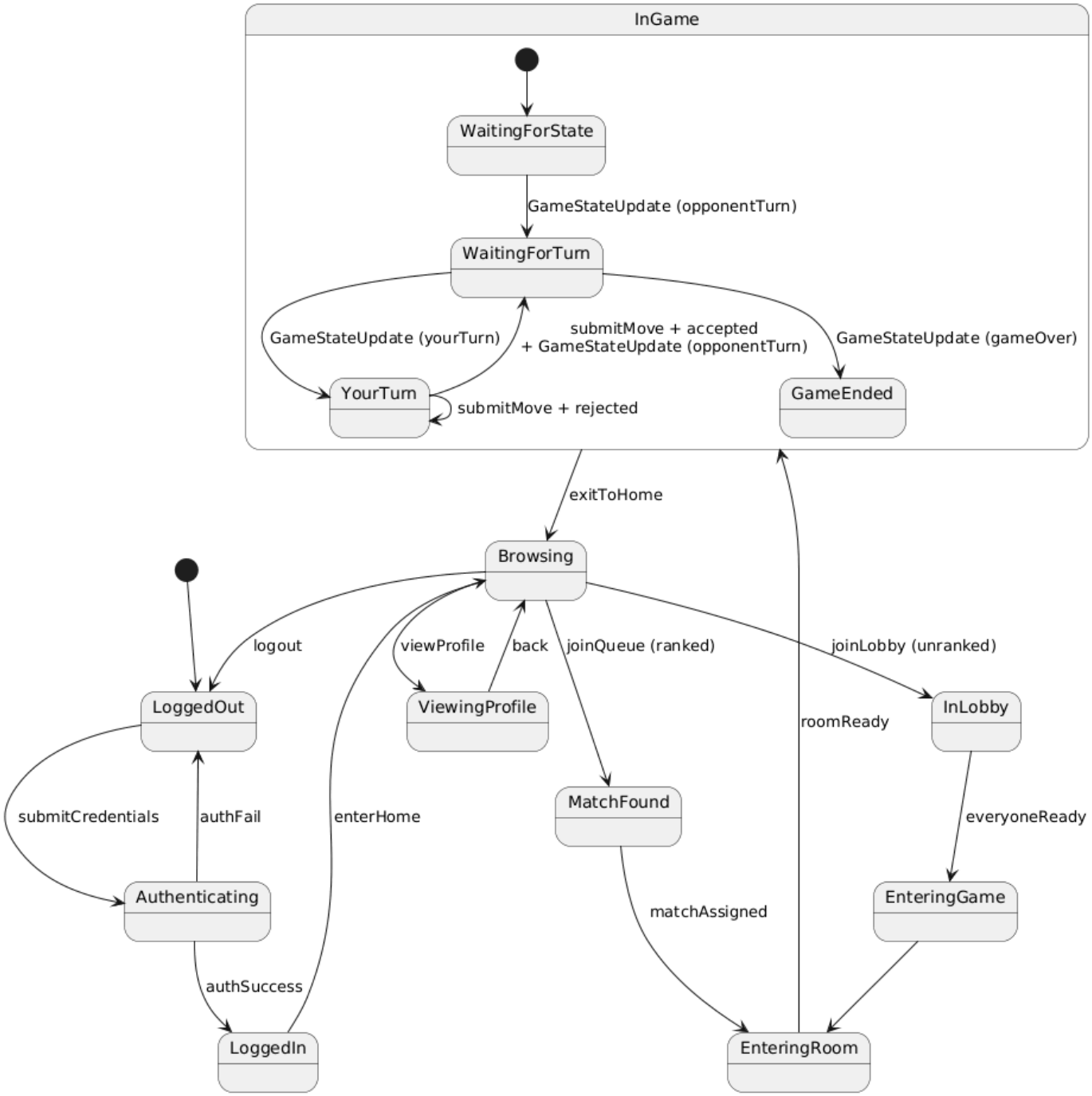
- `usernameLookup`, `addNewPlayer`, `deleteExistingPlayer`
- `findPlayerByUsername`, `saveDatabase(filename)`, `loadDatabase(filename)`

> Plan: HashMap or file load/unload. No external DB required for the stub.

5. State Machines

5.1 Player Session State Machine (Platform)

This machine tracks a single player's lifecycle across the entire platform.



Key constraints:

- Logout is only permitted from `Browsing` — never mid-game
- Ranked matches: player is auto-joined on `matchAssigned`; no option to decline
- Unranked matches: player joins a `Lobby`; can chat while waiting for others to ready up
- `PlatformCore` can send notifications to a player regardless of their current state

5.2 Game Session State Machine

This machine tracks a single game session (game-agnostic).

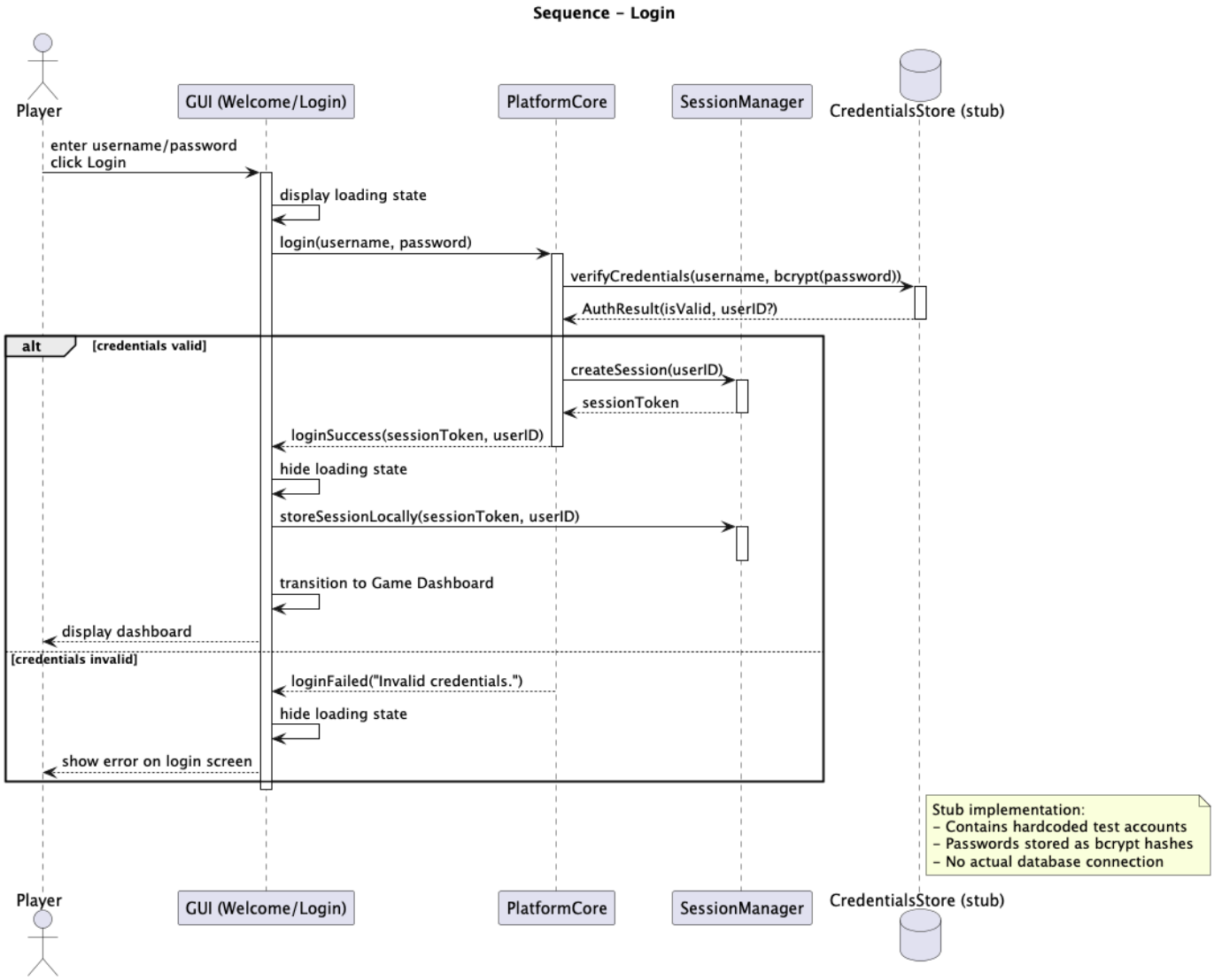
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Every transition emits a `GameStateUpdate` **to Core containing:** `activePlayerId` , `allowedActions` , board snapshot (or delta), and `status` (`YOUR_TURN / WAITING / ENDED`).

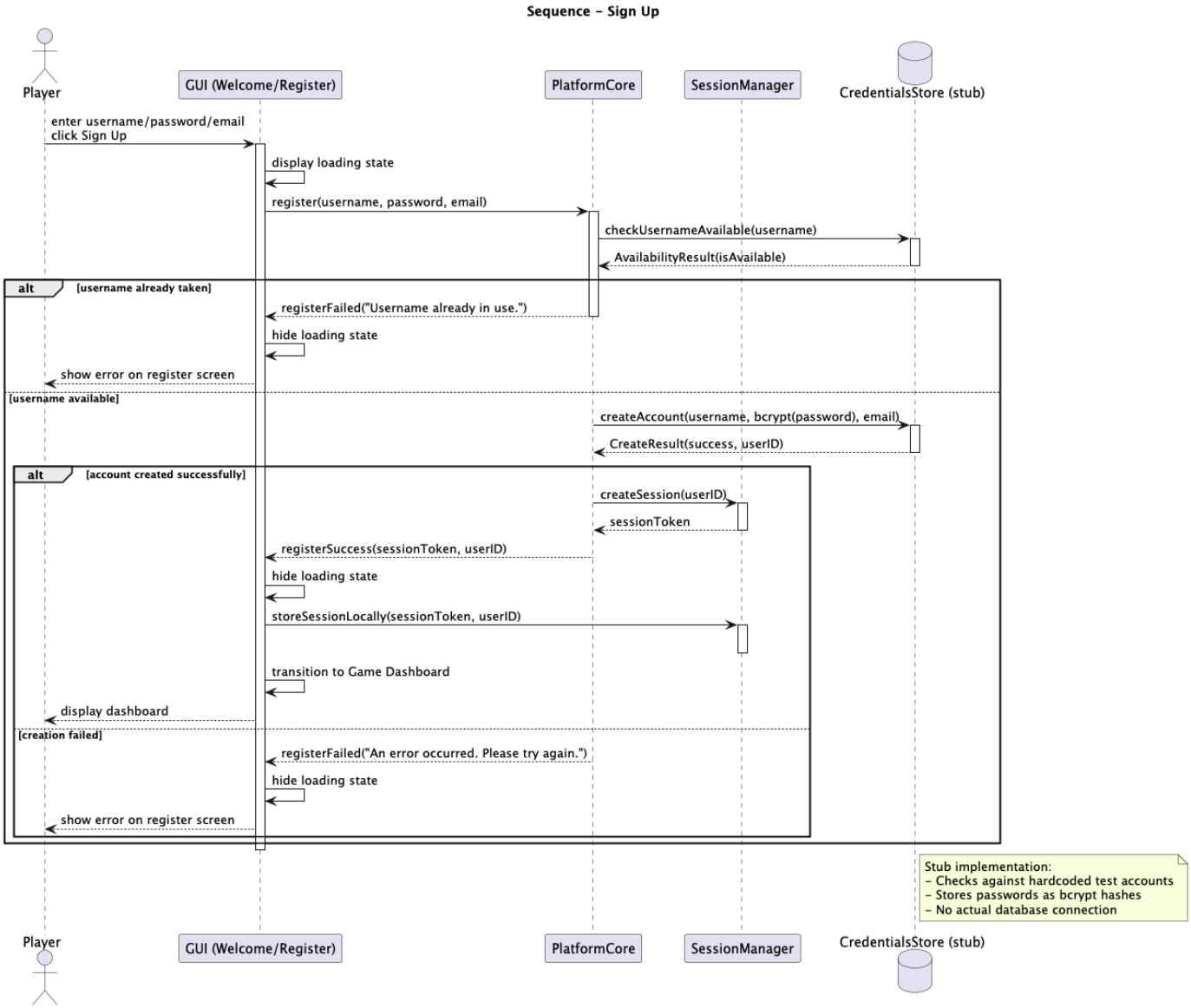
6. Key Sequence Flows

> **How to insert images:** In Markdown, use the syntax `![Alt text](path/to/image.png)` . For a local file in the same folder as this document, use a relative path like `![Login Sequence](images/04_01_01_Login_Sequence.png)` . For a file in the same directory, just `![Login Sequence](04_01_01_Login_Sequence.png)` works too. The alt text (inside `[ ]`) is shown if the image fails to load and is good practice for accessibility.

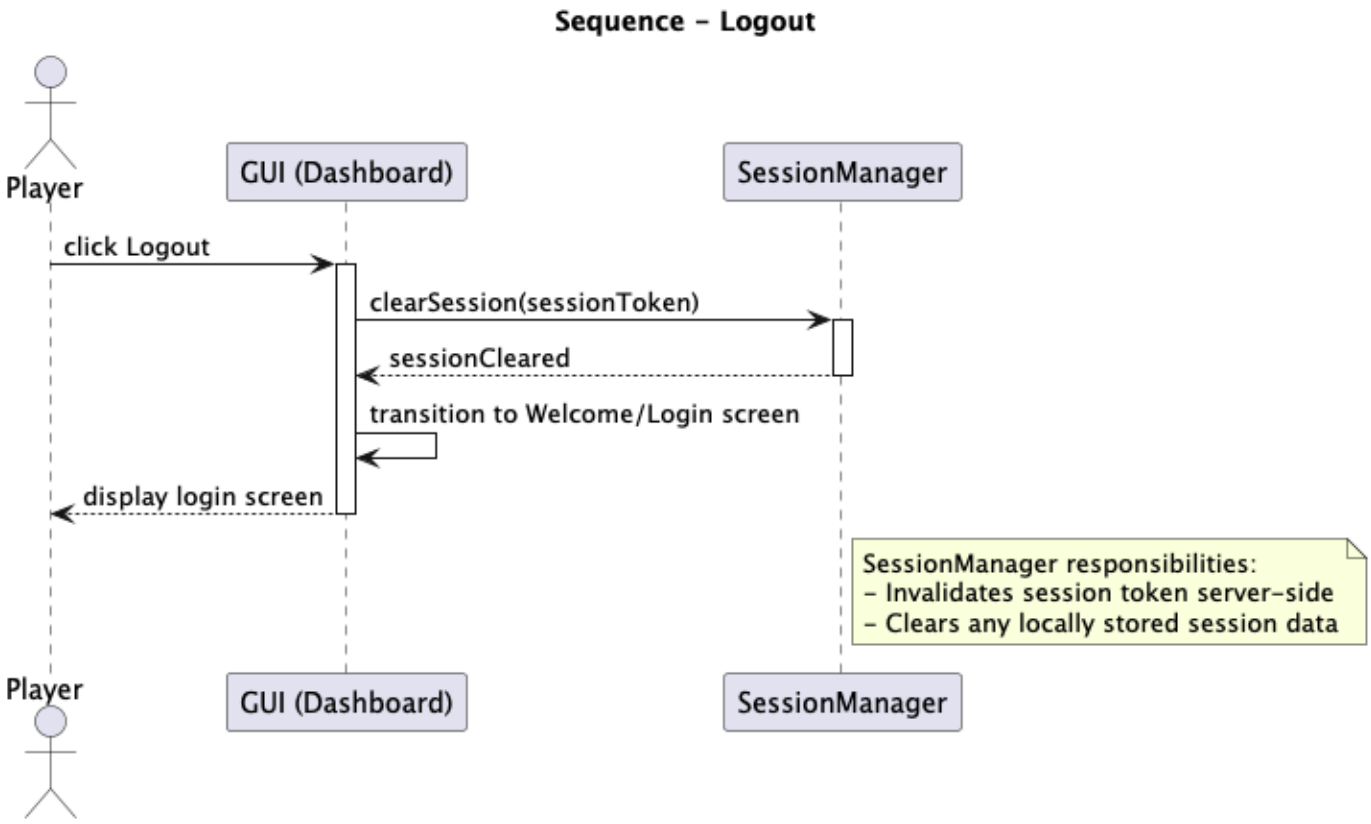
6.1 Login



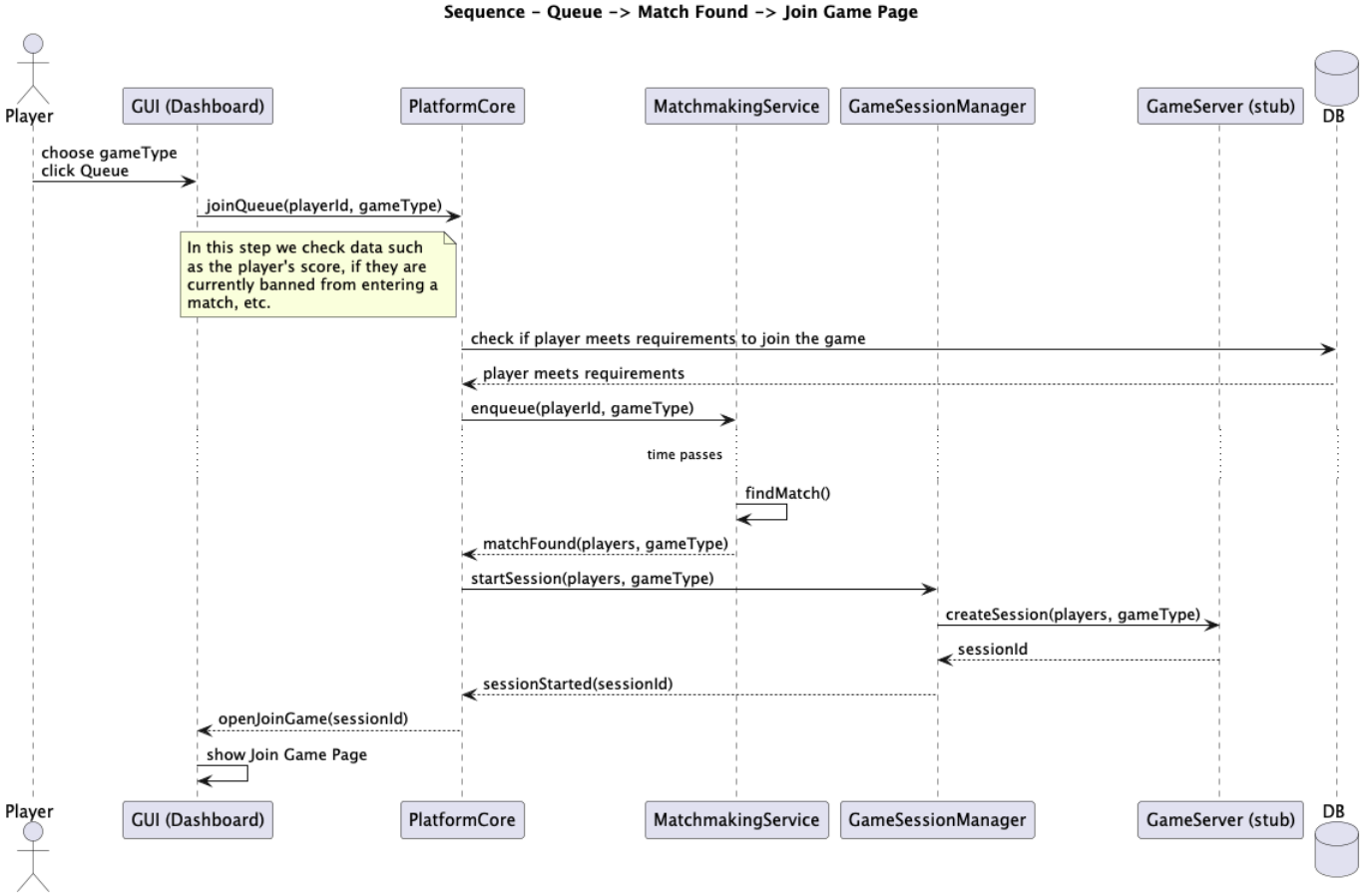
6.2 Sign Up



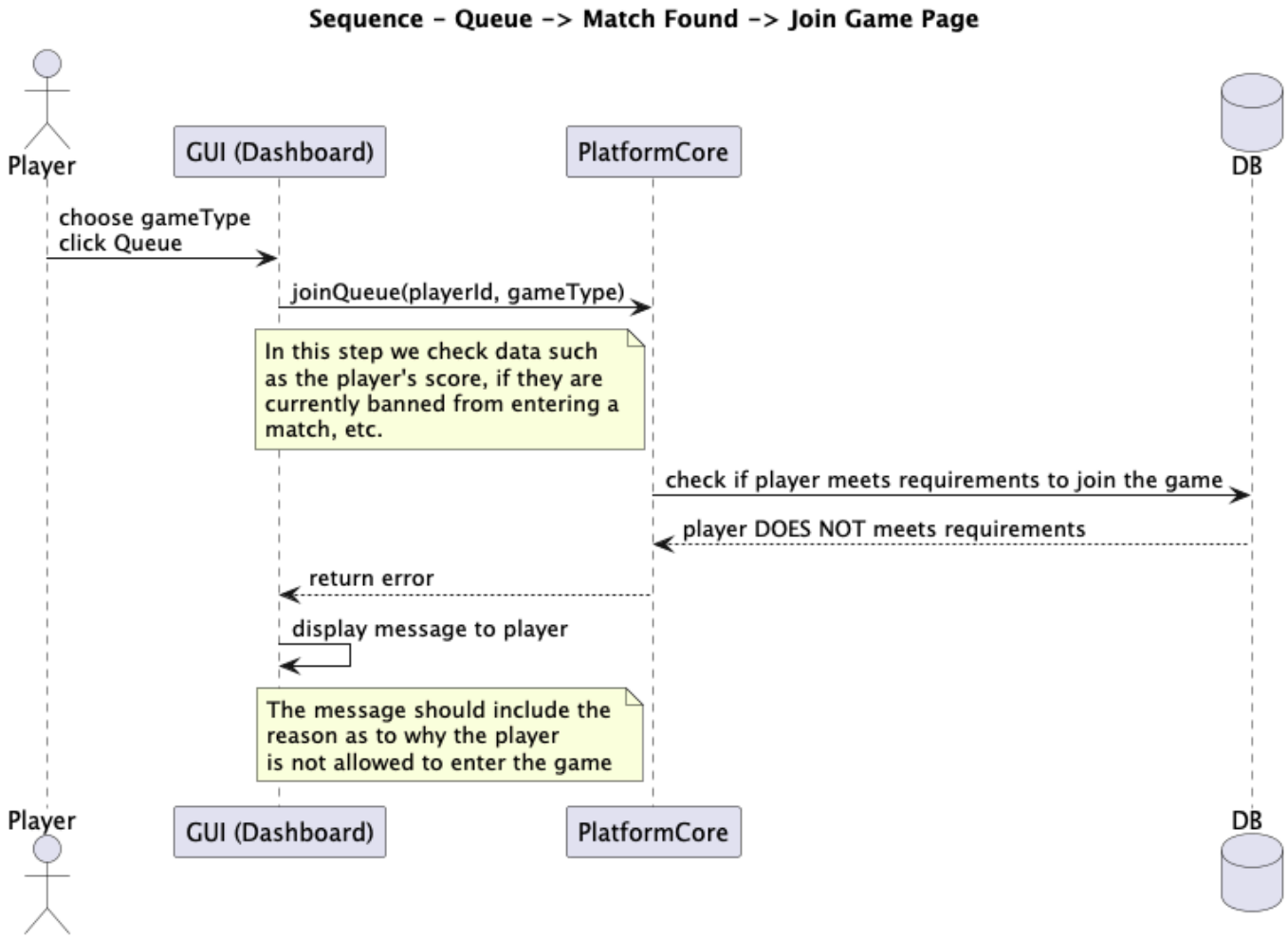
6.3 Logout



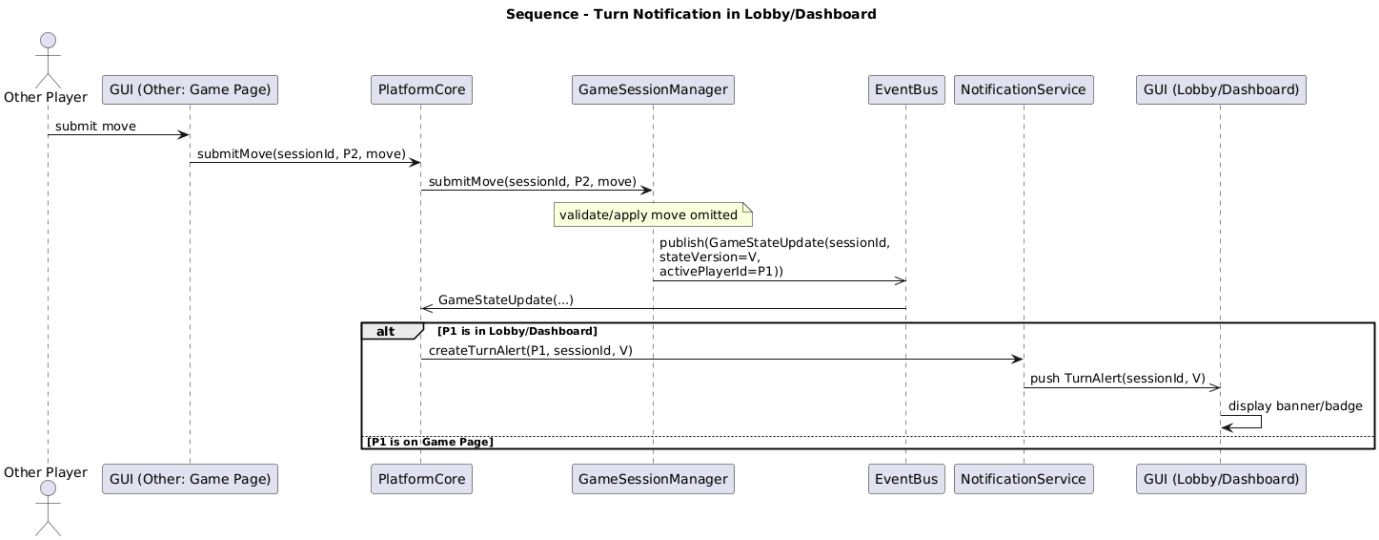
6.4 Join Game (Requirements Met)



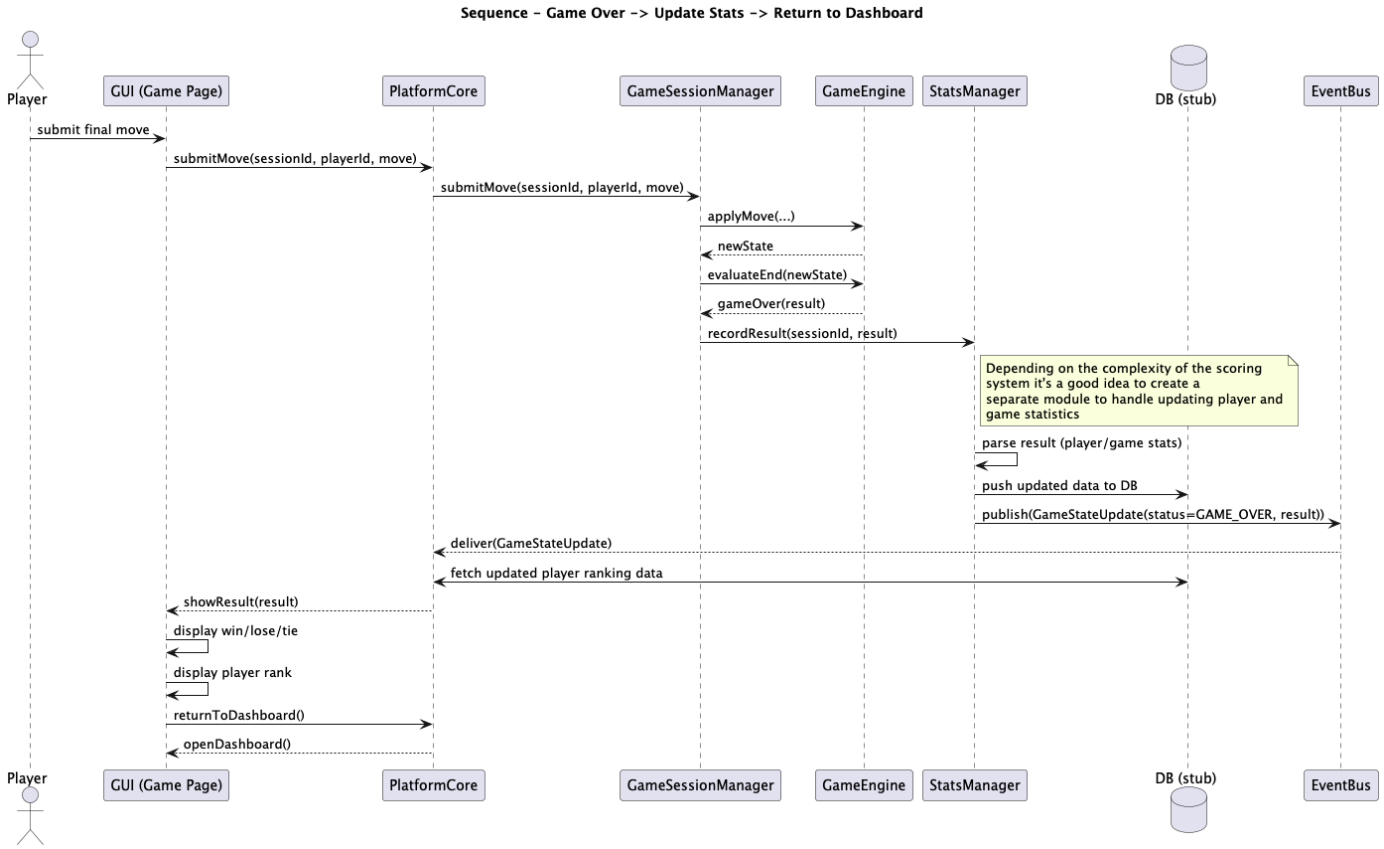
6.5 Join Game (Requirements Not Met)



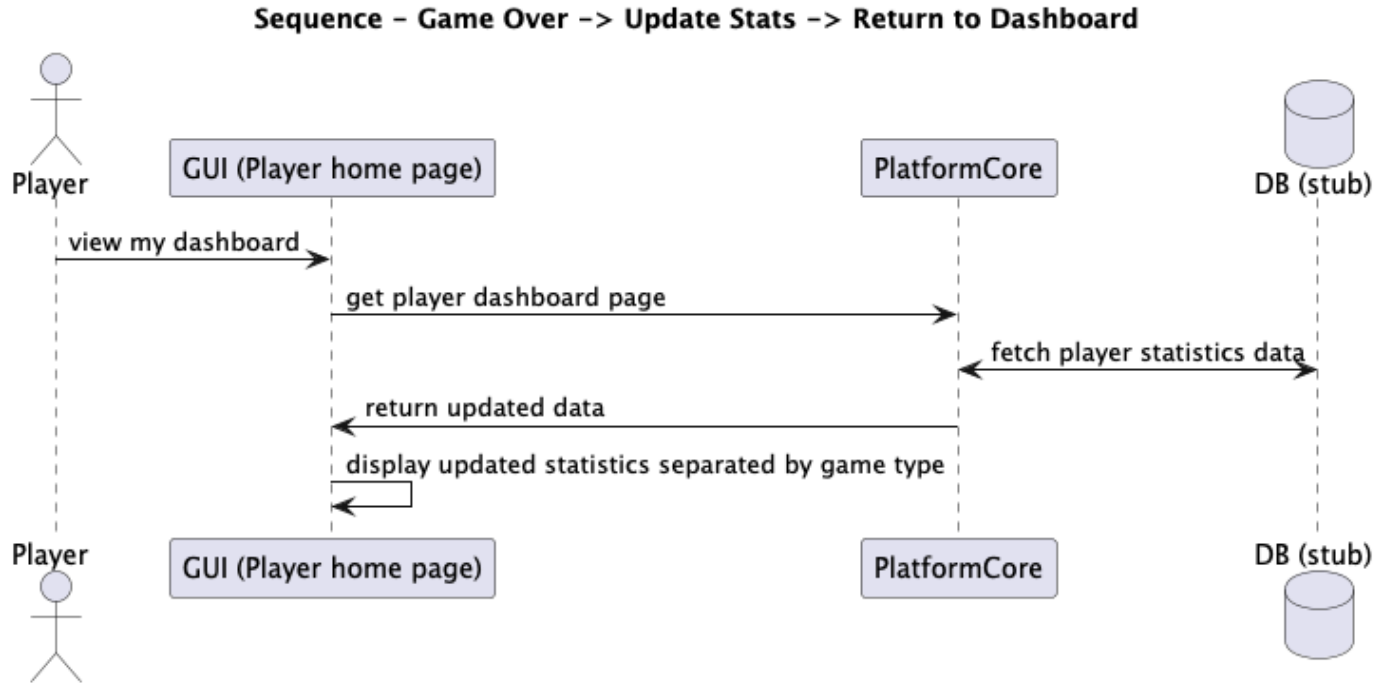
6.6 Turn Notification in Lobby/Dashboard



6.7 Game Over → Update Stats → Return to Dashboard



6.8 View Player Statistics



7. Cross-Cutting Concerns

Notification Routing

`PlatformCore` is the single authority for routing notifications. After receiving a `GameStateUpdate` from the `EventBus`, it inspects each recipient's current UI state:

- If the recipient is on the **game screen** → deliver board/turn update directly to `PlayGame`
- If the recipient is in the **lobby/dashboard** → deliver a turn alert banner/badge via `PlatformCore.notifyPlayer()`

This ensures turn notifications reach players regardless of where they are in the application.

Move Pipeline

All move submissions pass through the same pipeline regardless of game type:

```
UI.submitMove()
  → PlatformCore.submitMove()
    → GameSessionManager.submitMove()
      → Game.validateAction()          [game-specific, returns reason on failure]
      → Game.handleAction()            [game-specific]
      → Game.isGameOver()              [game-specific]
      → publish(GameStateUpdate)       [platform-level]
        → EventBus → PlatformCore → notify recipients
```

Authentication & Session Management

All authentication flows use bcrypt-hashed passwords. Session tokens are issued by `SessionManager` on successful login/registration and invalidated on logout. The `CredentialsDatabase` stub uses an in-memory `HashMap`; the real DB can be swapped in at that boundary.

Rank & MMR

`GeneralStats.updateMMR(win)` and `updateRank()` are called by `StatsManager` (invoked by `GameSessionManager`) after every game ends. The `Rank` enum defines seven tiers (Bronze → Grandmaster). `Leaderboard.updatePlayer()` reflects changes globally.

8. Stub & Integration Points

The following components are stubs with clearly defined interfaces, ready for replacement with real implementations:

Stub	Interface Contract	Replaces
`CredentialsDatabase`	`addNewPlayer`, `usernameLookup`, `findPlayerByUsername`, `saveDatabase` / `loadDatabase`	User authentication database
`Database`	`storeData()`, `retrieveData()`	Game data storage (match history, leaderboard, stats)
`Networking`	`handleConnections`, `syncPlayers`, `sendPlayerActions`, `disconnectPlayer`, `reconnectToGame`	Cloud hosting / server infrastructure
`GameServer`	`createSession(players, gameType): sessionId`	Online game server / cloud hosting

All stubs are marked with the `«stub»` stereotype in the class diagrams and contain hardcoded test data for local execution.

9. Technology Stack

Concern	Technology
Language	Java (JDK)
IDE	IntelliJ IDEA
UI Framework	JavaFX, Swing, or FXGL
Version Control	GitLab (csgit.ucalgary.ca)
Build / Run	Standard Java build (instructions in README)
Documentation	Markdown Files, PDF, PNG diagrams
Database (stub)	In-memory `HashMap` / file load-unload

Document version: aligned with design artifacts v1.0.0 (Iteration 1). This document is a living artifact and will be updated as implementation progresses through Iterations 2 and 3.